Citation Equity: A Writing Tool for AI Search That Tests Its Own Advice

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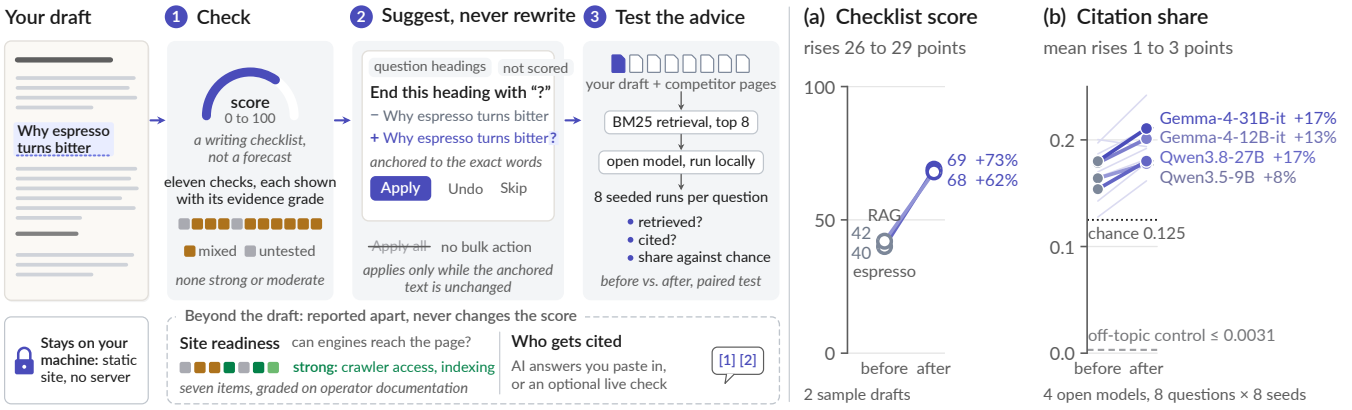


Figure 1: On two sample drafts, following the tool's suggestions raised its checklist score by 26 to 29 points but moved mean citation share on four open models by only 1 to 3 points. (a) Checklist score (the rise is built in). (b) Citation share in the sensitivity test.

ABSTRACT

AI answer engines now choose which pages a reader sees cited, yet writers get optimization advice with thin evidence, from tools that track brands or rewrite their text. We present Citation Equity, an open-source writing workspace that runs in the browser. It shows the graded evidence behind each of its eleven checks, suggests fixes anchored to exact passages that the author applies one at a time, and includes a citation test that separates being retrieved from being cited. Because any null finding counts against a check, none is strongly supported, and the tool calls its score a writing checklist, not a forecast. On our two sample drafts, following its suggestions raised that score from 40 to 69 and from 42 to 68 but moved mean citation share on four open models by only 1 to 3 points. Tested one change at a time on real web pages, only adding the reader's own words near the top clearly raised citation, by about 0.03 on a 0-to-1 scale, and kept its sign with another editor and other judges; question headings, cited sources, direct answer first and lists showed no effect larger than 0.03, and attributed quotes were inconclusive. The tool now scores what helped and marks the rest as unscored notes. All tests use open models given fixed sources, not live engines. At the venue, visitors run the loop on an example or their own text on one offline laptop.

CCS CONCEPTS

• **Information systems** → **Web searching and information discovery**; • **Human-centered computing** → *Interactive systems and tools.*

KEYWORDS

generative engine optimization, answer engines, citation, writing tools, evidence grading, null results

1 INTRODUCTION

Answer engines, systems that answer a question in prose and cite web sources, now sit between writers and readers. In a preregistered field experiment, a search page with only AI Mode cut click-through by 18.8 percentage points [13]. Generative engine optimization (GEO), changing a page so that answer engines cite it, now reaches writers as advice and as products.

The advice rests on weak evidence. Its best-known figure, gains of up to 40%, comes from one setup with five sources in

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one model's prompt [2], and a survey of 45 studies rejects it as a general claim [9]; adding statistics lowered citation rank in 19 of 24 C-SEO Bench settings [10]. Of 24 commercial products whose public pages we read on 4 October 2026, most sell brand tracking and seven also generate or rewrite content,[1] yet writers feel less ownership of text an AI writes for them [4, 5].

We present *Citation Equity*, an open-source writing workspace that runs in the browser (Figure 1).[2] It grades a draft on eleven checks with the evidence behind each, suggests anchored fixes the author applies one at a time, never rewrites on its own, and tests whether following the advice changed citation. Evidence tiers [1] and a readiness score kept apart from measured citation [6] exist in open tools; what is new is the combination, in one serverless author-side tool, of graded evidence under an explicit rule for nulls, a self-test of the advice, and a check of that test's sensitivity against targets set in advance.

Applying all the suggestions to two sample drafts raises the checklist score by 29 and 26 points, as built in. Tested one change at a time on real pages, only adding the reader's own words near the top clearly helped, by about 0.03 on a 0-to-1 citation score, keeping its sign with another editor and other judges; question headings, cited sources, direct answer first and lists showed no effect larger than 0.03 and attributed quotes were inconclusive, so the tool now scores what helped and marks what did not (Figure 5).

2 CITATION EQUITY

A writer pastes a draft, sets up to eight target questions, applies fixes one at a time and tests the revision. Measured outcomes are reported beside the score, never inside it, and drafts stay local unless the author picks a hosted model. A TypeScript monorepo holds an *engine* of pure check functions (browser and CLI), a *bench* with the citation test, and a *workspace*, a Next.js static export on GitHub Pages with no server; a browser test asserts no request leaves the origin on load. The engine checks a 1,000-word draft in about 10 ms on a laptop; 1,248 unit and 78 browser tests cover the code.

2.1 Check: eleven checks with graded evidence

Six checks come from the project's initial design, four from the literature (Figure 3 names them all), and one, *your reader's words up top*, from our own test, where it helped (Section 3); four (question headings, lists, cited sources, direct answer first) are shown and still suggest fixes but are not scored, as our tests found no clear effect on citation. Each check scores the draft from 0 to 1 and carries an evidence grade (strong, moderate, mixed or untested) under a written rubric in which a null or negative result in any study, even in one subgroup, makes a check mixed. Over the nine sources the tool cites, each opened and read, no check is strong or moderate: nine are mixed and two untested (Figure 3). A looser rule, defined post hoc, that counts only a study's own no-effect conclusion or a significant harm makes five mixed checks moderate. **[OWNER:** Second rater (Ishan Gupta): agreement pending.] The *checklist score* (the tool's AI-ready score) averages the checks out of 100 under editable weights and is labelled a writing checklist, not a forecast.

2.2 Suggest, never rewrite

Comments anchored to spans of text improved writers' agency over a chat interface in a lab study [8], so every suggestion is anchored to an exact range of the draft and follows the author's own edits. An *edit* (say, a missing question mark) is previewed, and Apply changes the range only if its text is unchanged; an *advisory* (say, name a real source for a figure) is left to the author. Every action has Undo, and there is no "apply all" (Figure 2). An optional local model can propose wording for one suggestion, shown side by side and applied only on the author's click. *Beyond the draft*, *site readiness* checks seven items of the published page (crawler access and indexing directives are graded strong, on operators' documentation); *who gets cited* sorts the sources of pasted AI answers; *your questions* turns keywords or Search Console queries into a ranked list of reader questions for the whole site, each with the searches an answer engine is likely to run behind it. None changes the score, and the last two are untested helpers.

2.3 Test it: the citation bench

For each question, BM25 retrieves the top 8 of the draft and its competitor pages, and a model the author runs (`llama3.2:3b` through Ollama by default) answers with numbered citations under GEO's prompt plus one format reminder [2], 8 times with different seeds and source orders [9, Sec. 11.2]. Reports keep apart how often the draft was retrieved, how many answers were *valid* (failures are outcomes, not discarded data), how often it was cited, and its *word share* beside the *chance level* of an even split (0.125), with a paired sign-flip test; each says *"It does not predict live AI engines."* For larger models it exports the prompts and replays answers generated elsewhere.

3 WHAT THE TESTS FOUND

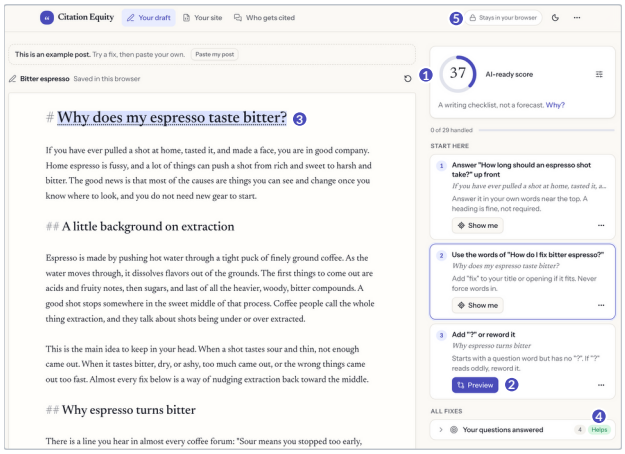
On the authors' laptop the bench shows a direction at most: every 95% interval of the draft's word share included the 0.125 chance level, $p \geq 0.125$ with 4 questions, and a first recording, misread through truncated prompts, was withdrawn. A sensitivity test planned in advance, with four open models and synthetic competitors, measured *citation share*: a one-sentence answer edit gained only +0.007 to +0.034 against a +0.10 target the harness could detect, and the full after drafts, 14% and 22% longer, gained +0.014 to +0.031 (Figure 1b).

One change at a time. Our first test applies one check's suggestion alone to real pages (80 informational GEO-bench queries [2], replicated on 60 from C-SEO Bench [10]): an open model edits only the spans the check flags, and each edit is compared with a same-length paraphrase of those spans; two open judges see the page among 4 sources, and its *citation score* is $(5 - r)/4$ when it is the r-th source cited, else 0 (plans fixed in advance, code and data: footnote 2).

Of single changes, only the two that add the reader's own words near the top clearly raised the score (Figure 4): *target questions answered*, done as a section headed by the question (+0.027, 95% CI

[1] For example https://www.evertune.ai/pricing and https://writesonic.com/pricing, opened 24 October 2026.

[2] Live demo: https://shehral.github.io/citation-equity/. Code (MIT): https://github.com/shehral/citation-equity. Data: https://huggingface.co/datasets/Shehral/citation-equity-experiments. Research page (plans, results): https://shehral.github.io/citation-equity/research/. Video: https://youtu.be/kbU9t3ewVUk.



1. AI-ready score: "A writing checklist, not a forecast."
2. The ready edit: Preview first, Apply only inside the preview
3. The selected fix's passage, marked in the draft
4. A check's tag from our tests: "Helps"
5. Privacy chip: "Stays in your browser"

Figure 2: The deployed draft view (espresso example): the author applies one fix at a time; a tag marks the checks our tests found to help.

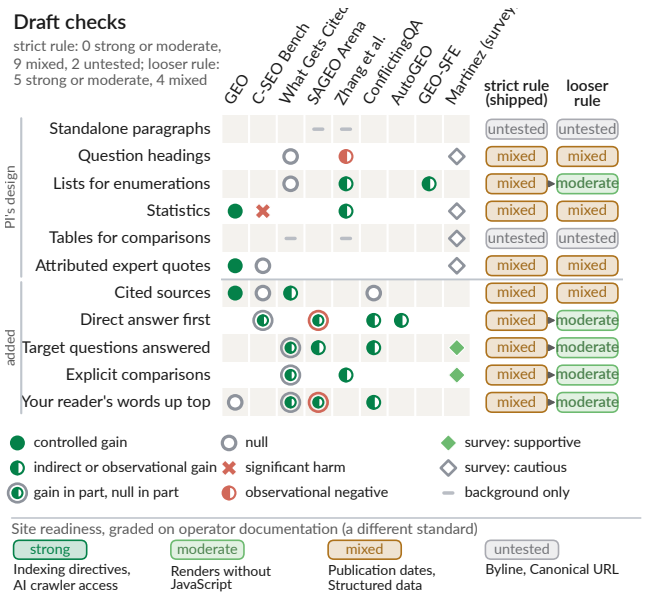


Figure 3: Every tested draft check has a null or negative result somewhere, but "mixed" spans positive-in-every-model to harmful, and a looser rule, defined post hoc, moves five to moderate. Columns: [2], [10], [11], [7], [16], [12], [14], [15], [9]. One rater.

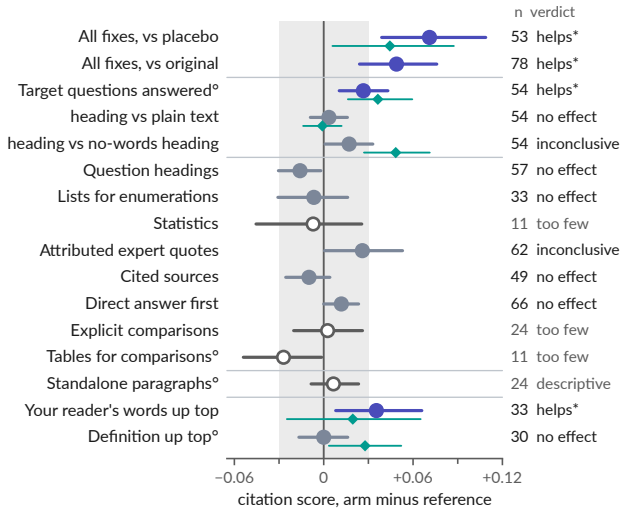


Figure 4: Of single changes, only the two adding the reader's words up top clearly raised citation; the heading form, sources and answer-first showed no effect larger than 0.03; quotes were inconclusive. Difference from a same-length paraphrase or the original (°), 95% intervals; filled: 30+ pages; band: ±0.03; teal: replication; helps*: possibly via question-word overlap.

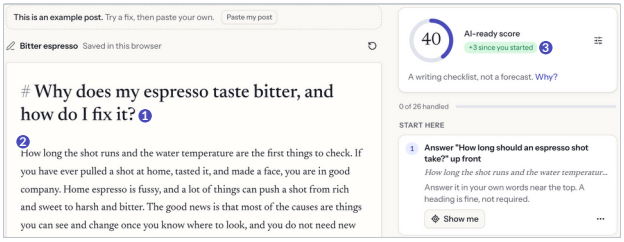
[+0.010, +0.043], Holm $p = 0.006$; replication +0.036), and *your reader's words up top*, the question's missing words in title and lead (+0.035); for both, the effect left after regressing out length and question-word changes includes zero. Follow-up tests, each planned before its answers, place the gain in the words, not the form: the question heading did no better than the same text set as plain (+0.003; replication −0.001) while the plain-text question scored higher than the untouched page (+0.023; +0.037). With another editor and other judges, the reader's-words gain kept its sign (+0.029, $n = 29$), as did all fixes together (+0.069, $n = 44$). A second follow-up added 180 queries for the checks with too few pages: over 260, question headings, cited sources and direct answer first showed no effect larger than 0.03 and attributed quotes were inconclusive; tables, statistics and comparisons still had too few.

4 RELATED WORK

The bench reuses GEO's prompt and word share [2] and the counterfactual style of C-SEO Bench and What Gets Cited, which edit one page and hold the rest fixed [10, 11]. Bajemon and Rochet validate a content score against engines through falsification gates [3] (a vendor-linked preprint); we run no such validation and do not claim our score predicts citation. Among open tools, aiseo-audit keeps an evidence tier for every factor and already demotes lists and tables to unscored diagnostics [1], and Citation Equity adapts code from it.

5 DEMONSTRATION AT THE VENUE

One laptop serves the static build and runs Ollama (`llama3.2:3b`), so the core demonstration needs no venue network. A second screen shows the hosted demo, and a poster carries Figures 1 and 4.



1 The word the first tip named ("fix"), typed into the title
2 A new opening sentence with the words of the other two questions
3 Score 37 to 40; rerun on the same edits, the engine moves only the reader's-words check

Figure 5: Following the reader's-words tips raises the score: the tool now scores what helped.

Station 1, the 90-second loop. The visitor loads the espresso example, follows the reader's-words advice and watches the score change (Figure 5), applies a ready edit through Preview (a heading's question mark, an unscored note, so the score holds), opens the studies behind one mixed check and the recorded citation test beside its chance line. A live run (64 answers, 15 minutes) starts each session and fills in while visitors watch. *Station 2, bring your own text (offline).* Visitors paste their own paragraph and see checks, suggestions and grades; the text never leaves the laptop. *Station 3, can engines reach your site? (online).* The command-line `ce site` runs the seven readiness items on the visitor's URL over a phone hotspot; offline, pasted HTML runs five of them in the browser.

6 ETHICS AND USE OF DATA

No human participants took part and no user data was collected. Sample drafts and competitor pages are invented; C-SEO Bench is downloaded at run time, never redistributed; building competitor pages from URLs respects robots.txt. On dual use, it generates no page text on its own, asks for real and checkable material, and states the FTC's endorsement rules; the bench could test a manipulative edit, but the tool offers none. The tests use the public pages GEO-bench lists [2]; their text, and everything built from it, is not redistributed, only URLs, hashes and results. A future user study will go through IRB review.

7 LIMITATIONS AND FUTURE WORK

Every measurement uses open models over fixed sources, not live engines, which differ and drift [9]; the sensitivity test covers two synthetic topics and 8 questions, the one-change tests 320 English queries, nearly all edited by one model family, with no positive control in the sensitivity design. The harness prompt has not been byte-checked against the bench's. Next are a comparison with live engines and a study with writers.

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Use of generative AI. We used Claude (Anthropic) as an assistant for software development, experiment and analysis code, and writing and editing. The authors designed the study, checked the results and take full responsibility for the paper.

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